

國立高雄大學 108 學年度轉學招生考試試題(轉二年級)

科目：微積分

系所：

電機工程學系(無組別)

是否使用計算機：是

考試時間：80 分鐘

本科原始成績：100 分

■ 以下考題總共十題，每題為十分。請根據題號依序作答。

1. Find $\lim_{x \rightarrow 0} \frac{\cos x(1 - \cos x)}{x}$.
2. Find $\frac{d}{dx} \left[\frac{\cos x}{1 + \cos x} \right]$.
3. Find $\frac{d}{dx} [(1 + \sec x)^{36}]$.
4. Find $\frac{d^2}{dx^2} [x^{1/3}]$.
5. $(x + 1)y^2 + \sin y = x^2$. Determine $\frac{dy}{dx}$.
6. Find $\int \sec^2 x (\tan x + 10)^{15} dx$.
7. Find $\int \left(\frac{a^x}{a^x + 1} \right) dx$ where a is a positive number and $a \neq 1$.
8. $z = (x + 1)^3 \sin y + y \cos x$. Calculate the total differential of z .
9. $f(x, y) = 3x + 5y$. Find $\nabla f(x, y)$.
10. Prove that $\vec{F}(x, y) = (3x^2 + 2)\vec{i} + 3y^2\vec{j}$ is conservative.

科目：物理

系所：電機工程學系(無組別)

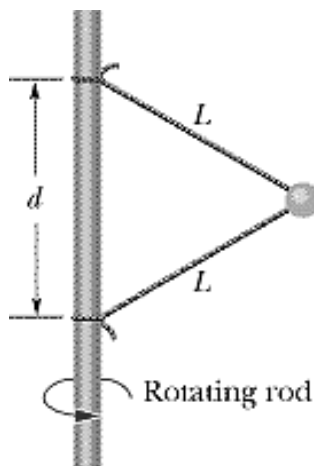
是否使用計算機：是

考試時間：80 分鐘

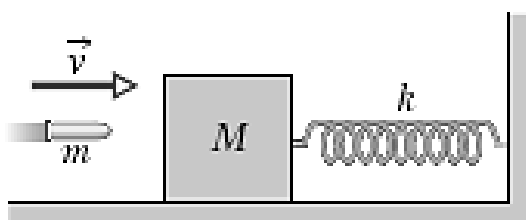
本科原始成績：100 分

*Gravitational acceleration, $g = 9.8 \text{ m/s}^2$

1. (10%) In the figure, a 1.34 kg ball is connected by means of two massless strings (無質量弦), each of length $L = 1.70 \text{ m}$, to a vertical, rotating rod. The strings are tied to the rod with separation $d = 1.70 \text{ m}$ and are taut (拉緊的). The tension in the upper string is 35.0 N . Please find the speed of the ball.



2. (5% + 5%) A block of mass $M = 5.4 \text{ kg}$, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant $k = 6000 \text{ N/m}$. A bullet of mass $m = 9.50 \text{ g}$ and velocity of magnitude 680 m/s strikes and is embedded in the block. Assuming the compression of the spring is negligible until the bullet is embedded. Please determine (a) the speed of the block immediately after the collision. (b) the amplitude of the resulting simple harmonic motion.



科目：物理

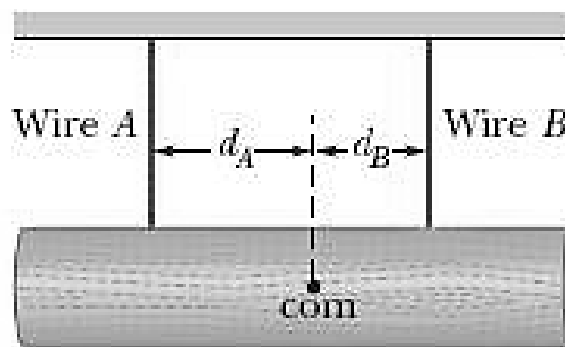
系所：電機工程學系(無組別)

是否使用計算機：是

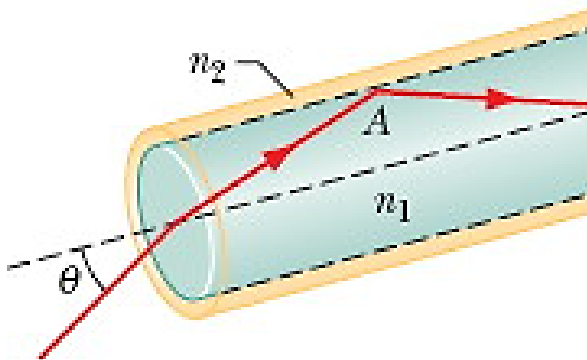
考試時間：80 分鐘

本科原始成績：100 分

3. (15%) In the figure, a 110 kg uniform log hangs by two steel wires, A and B, both of radius 1.15 mm . Initially, wire A was 2.40 m long and 1.85 mm shorter than wire B. The log is now horizontal. Young's modulus for steel is $2.00 \times 10^{11} \text{ N/m}^2$. Please find the ratio of d_A/d_B .



4. (15%) The figure depicts a simplistic optical fiber: a plastic core ($n_1 = 1.58$) is surrounded by a plastic sheath ($n_2 = 1.46$). A light ray is incident on one end of the fiber at angle θ . The ray is to undergo total internal reflection at point A, where it encounters the core-sheath boundary. (Thus, there is no loss of light through that boundary.) What is the maximum value of θ that allows total internal reflection at A?



θ (degree)	$\sin(\theta)$	θ (degree)	$\sin(\theta)$
30.0	0.5000	35.5	0.5807
30.5	0.5075	36.0	0.5878
31.0	0.5150	36.5	0.5948
31.5	0.5225	37.0	0.6018
32.0	0.5299	37.5	0.6088
32.5	0.5373	38.0	0.6157
33.0	0.5446	38.5	0.6225
33.5	0.5519	39.0	0.6293
34.0	0.5592	39.5	0.6361
34.5	0.5664	40.0	0.6428
35.0	0.5736	40.5	0.6494

科目：物理

系所：電機工程學系(無組別)

是否使用計算機：是

考試時間：80 分鐘

本科原始成績：100 分

5. (15%) The circuit in the figure consists of switch “S”, a 12 V ideal battery, a $20\text{ M}\Omega$ resistor (R), and an air-filled capacitor (C). The capacitor has parallel circular plates of radius 6 cm, separated by 3 mm. At time $t = 0$, switch “S” is closed to begin charging the capacitor. The electric field between the plates is uniform. At time $t = 250\text{ }\mu\text{s}$, what is the magnitude of the magnetic field within the capacitor, at radial distance, $r = 3\text{ cm}$? (Note: vacuum permeability, $\mu_0 = 4\pi \times 10^{-7}\text{ H/m}$)



6. (10% + 5%) $R_1 = 8.0\text{ }\Omega$, $R_2 = 10.0\text{ }\Omega$, $L_1 = 0.3\text{ H}$, $L_2 = 0.2\text{ H}$, and the ideal battery has $\mathcal{E} = 6.0\text{ V}$ as shown. **(a)** Just after switch “S” is closed, determine what rate of current changing is in inductor 1 (L_1)? **(b)** When the circuit is in the steady state, what is the current in inductor 1?

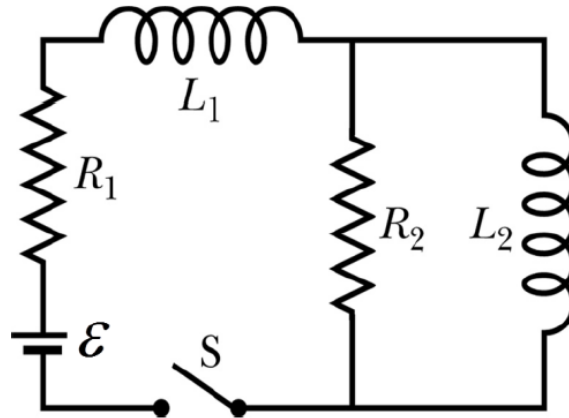
科目：物理

系所：電機工程學系(無組別)

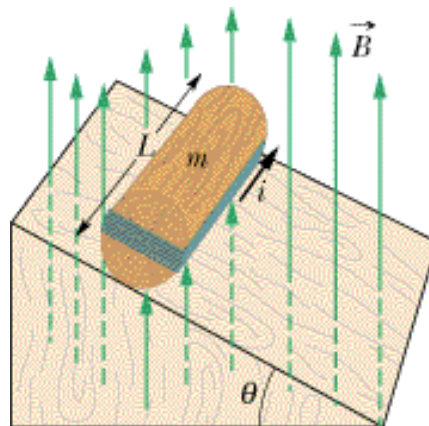
是否使用計算機：是

考試時間：80 分鐘

本科原始成績：100 分



7. (10%) The figure shows a wood cylinder of mass $m = 0.15 \text{ kg}$ and length $L = 0.1 \text{ m}$, with $N = 13$ turns of wire wrapped around it longitudinally, so that the plane of the wire coil contains the long central axis of the cylinder. The cylinder is released on a plane inclined at an angle θ to the horizontal, with the plane of the coil parallel to the incline plane. If there is a vertical uniform magnetic field of magnitude 0.92 T , what is the least current (i) through the coil that keeps the cylinder from rolling down the plane?



8. (10%) A capacitor in an Inductor-Capacitor (LC) circuit oscillator has a maximum potential difference 15V and a maximum energy of $360 \mu\text{J}$. At a certain instant the energy in the capacitor is $40 \mu\text{J}$. At that instant what is the

國立高雄大學 108 學年度轉學招生考試試題

科目：物理

系所：電機工程學系(無組別)

是否使用計算機：是

考試時間：80 分鐘

本科原始成績：100 分

Electromotive force (emf) induced in the inductor?